

Semester	B.E. Semester VII – Computer Engineering
Subject	R Programming
Subject Professor In- charge	Prof. Sanjeev Dwivedi
Assisting Teachers	Prof. Sanjeev Dwivedi
Laboratory	M-411B

Student Name	Sneha Gadhari
Roll Number	23102B0055
Grade and Subject Teacher's Signature	

Experiment Number	03
Experiment Title	Multi-source Retail Sales Data Integration and Analysis

Resources / Apparatus Required	Hardware: Computer system	Software: Colab
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Objective	The objective of this lab is to design and implement an R-based data pipeline that imports retail business data from three heterogeneous sources — CSV (transactions), JSON (products), and Excel (customers) — cleans and integrates them using dplyr, and analyzes the combined dataset to generate actionable sales and customer insights, with the final processed dataset persisted to a SQLite database for future querying.
Description	This experiment simulates a real-world retail analytics scenario where a multinational company's data is fragmented across different systems and formats. The UCI Online Retail dataset (541,909 transaction records) was used as the source, split programmatically into transactions.csv, products.json, and customers.xlsx to mirror the assignment's multi-source setup. The workflow followed four stages: • Import & Clean (Task 1): Data was read using readr, jsonlite, and readxl. Cleaning removed 5,429 duplicate rows, 135,080 rows missing CustomerID, all negative/zero-quantity rows (returns), and 215 products with invalid (≤ 0) prices. A Revenue = Quantity $\times$ UnitPrice field was engineered after joining valid prices, leaving 387,877 usable rows. • Integrate (Task 2): inner_join() was used to attach UnitPrice (since a transaction without a valid price can't generate revenue), while left_join() was used to attach Description and Country so no valid cleaned transaction is lost even if enrichment data is missing. Post-join checks confirmed zero unmatched records for description or country. • Analyze (Task 3): Using group_by()/summarise(), the analysis computed total revenue (£10,752,840), top 5 products/countries/customers by revenue, and a case_when()-based customer segmentation into Low/Medium/High/Premium tiers using spend quartiles (25th/50th/75th percentile thresholds). The UK was identified as the dominant market and Saudi Arabia as the weakest. • Store & Query (Task 4): The final integrated table (retail_data, 387,877 rows $\times$ 9 columns) was written to a SQLite database (retail_analysis.db) as table retail_sales using DBI/RSQLite, and queried directly with SQL (GROUP BY + ORDER BY) to reproduce the top-5-

customers and revenue-by-country results, confirming consistency between the dplyr and SQL analysis paths.

Output

Downloaded raw dataset -> 541909 rows, 8 columns

A tibble: 6 × 8

InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
<chr>	<chr>	<chr>	<dbl>	<dtm>	<dbl>	<dbl>	<chr>
536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850	United Kingdom
536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850	United Kingdom
536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
536365	22752	SET 7 BABUSHKA NESTING BOXES	2	2010-12-01 08:26:00	7.65	17850	United Kingdom

transactions.csv : 541909 rows
products.json : 4070 unique SKUs
customers.xlsx : 4372 unique customers

Rows: 541,909
Columns: 5
\$ InvoiceNo <chr> "536365", "536365", "536365", "536365", "536365", "536365" ...
\$ StockCode <chr> "85123A", "71053", "84406B", "84029G", "84029E", "22752", ...
\$ CustomerID <dbl> 17850, 17850, 17850, 17850, 17850, 17850, 17850, 17850, 17...
\$ Quantity <dbl> 6, 6, 8, 6, 6, 2, 6, 6, 6, 32, 6, 6, 8, 6, 6, 3, 2, 3, 3, ...
\$ InvoiceDate <dtm> 2010-12-01 08:26:00, 2010-12-01 08:26:00, 2010-12-01 08:2...
Rows: 4,070
Columns: 3
\$ StockCode <chr> "85123A", "71053", "84406B", "84029G", "84029E", "22752", ...
\$ Description <chr> "WHITE HANGING HEART T-LIGHT HOLDER", "WHITE METAL LANTERN...
\$ UnitPrice <dbl> 2.55, 3.39, 2.75, 3.39, 3.39, 7.65, 4.25, 1.85, 1.85, 1.69...
Rows: 4,372
Columns: 2
\$ CustomerID <dbl> 17850, 13047, 12583, 13748, 15100, 15291, 14688, 17809, 153...
\$ Country <chr> "United Kingdom", "United Kingdom", "France", "United Kingd...

Dimensions -> transactions: 541909 5 | products: 4070 3 | customers: 4372 2

Missing values (transactions):
InvoiceNo StockCode CustomerID Quantity InvoiceDate
0 0 135080 0 0

Missing values (products):
StockCode Description UnitPrice
0 176 0

Missing values (customers):
CustomerID Country
0 0

Duplicate rows in transactions: 5429
Zero-quantity rows: 0
Negative-quantity rows (returns/cancellations): 10624
Zero/negative-price products: 215

Rows retained after attaching a valid UnitPrice: 387877
A tibble: 5 × 7

InvoiceNo	StockCode	CustomerID	Quantity	InvoiceDate	UnitPrice	Revenue
<chr>	<chr>	<dbl>	<dbl>	<dtm>	<dbl>	<dbl>
536365	85123A	17850	6	2010-12-01 08:26:00	2.55	15.30
536365	71053	17850	6	2010-12-01 08:26:00	3.39	20.34
536365	84406B	17850	8	2010-12-01 08:26:00	2.75	22.00
536365	84029G	17850	6	2010-12-01 08:26:00	3.39	20.34
536365	84029E	17850	6	2010-12-01 08:26:00	3.39	20.34

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InvoiceNo	StockCode	CustomerID	Quantity	InvoiceDate	UnitPrice	Revenue	Description	Country
<chr>	<chr>	<dbl>	<dbl>	<dtm>	<dbl>	<dbl>	<chr>	<chr>
536365	85123A	17850	6	2010-12-01 08:26:00	2.55	15.30	WHITE HANGING HEART T-LIGHT HOLDER	United Kingdom
536365	71053	17850	6	2010-12-01 08:26:00	3.39	20.34	WHITE METAL LANTERN	United Kingdom
536365	84406B	17850	8	2010-12-01 08:26:00	2.75	22.00	CREAM CUPID HEARTS COAT HANGER	United Kingdom
536365	84029G	17850	6	2010-12-01 08:26:00	3.39	20.34	KNITTED UNION FLAG HOT WATER BOTTLE	United Kingdom
536365	84029E	17850	6	2010-12-01 08:26:00	3.39	20.34	RED WOOLLY HOTTIE WHITE HEART.	United Kingdom
536365	22752	17850	2	2010-12-01 08:26:00	7.65	15.30	SET 7 BABUSHKA NESTING BOXES	United Kingdom

transactions_clean rows: 387877
retail_data rows : 387877

Rows with no matching product description: 0
Rows with no matching customer/country : 0

A tibble: 5 × 3

StockCode	Description	TotalRevenue
<chr>	<chr>	<dbl>
23843	PAPER CRAFT , LITTLE BIRDIE	168469.60
47566	PARTY BUNTING	142437.56
22423	REGENCY CAKESTAND 3 TIER	135604.80
85123A	WHITE HANGING HEART T-LIGHT HOLDER	93745.65
23166	MEDIUM CERAMIC TOP STORAGE JAR	81032.64

25% 50% 75%
361.005 802.600 2000.640

High Value Low Value Medium Value Premium
1084 1085 1085 1085

High-performing market: United Kingdom | Revenue: £ 8,861,857 | Customers: 3921
Underperforming market: Saudi Arabia | Revenue: £ 181.44 | Customers: 1

	A data.frame: 5 × 2 <table><tr><th>CustomerID</th><th>TotalRevenue</th></tr><tr><th><dbl></th><th><dbl></th></tr><tr><td>18102</td><td>408760.0</td></tr><tr><td>14646</td><td>357531.1</td></tr><tr><td>17450</td><td>186038.0</td></tr><tr><td>14911</td><td>182690.5</td></tr><tr><td>16446</td><td>168472.5</td></tr></table> A data.frame: 10 × 2 <table><tr><th></th><th>Country</th><th>TotalRevenue</th></tr><tr><th></th><th><chr></th><th><dbl></th></tr><tr><td>1</td><td>United Kingdom</td><td>8861857.13</td></tr><tr><td>2</td><td>Netherlands</td><td>363884.48</td></tr><tr><td>3</td><td>EIRE</td><td>331660.17</td></tr><tr><td>4</td><td>Germany</td><td>263818.97</td></tr><tr><td>5</td><td>France</td><td>226975.60</td></tr><tr><td>6</td><td>Australia</td><td>173918.61</td></tr><tr><td>7</td><td>Spain</td><td>67426.09</td></tr><tr><td>8</td><td>Switzerland</td><td>66619.97</td></tr><tr><td>9</td><td>Japan</td><td>48600.22</td></tr><tr><td>10</td><td>Belgium</td><td>47858.02</td></tr></table>	CustomerID	TotalRevenue	<dbl>	<dbl>	18102	408760.0	14646	357531.1	17450	186038.0	14911	182690.5	16446	168472.5		Country	TotalRevenue		<chr>	<dbl>	1	United Kingdom	8861857.13	2	Netherlands	363884.48	3	EIRE	331660.17	4	Germany	263818.97	5	France	226975.60	6	Australia	173918.61	7	Spain	67426.09	8	Switzerland	66619.97	9	Japan	48600.22	10	Belgium	47858.02
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Conclusion	The lab successfully demonstrates an end-to-end R pipeline for integrating and analyzing multi-format retail data, revealing that revenue and customer value are both heavily concentrated (UK ≈ 82% of revenue; a handful of customers and SKUs driving outsized returns), while also surfacing meaningful data-quality issues in the raw feed that would need upstream fixes in a production setting. The exercise reinforced practical skills in dplyr joins, case_when() segmentation, and SQLite integration for reproducible business analytics in R.																																																		